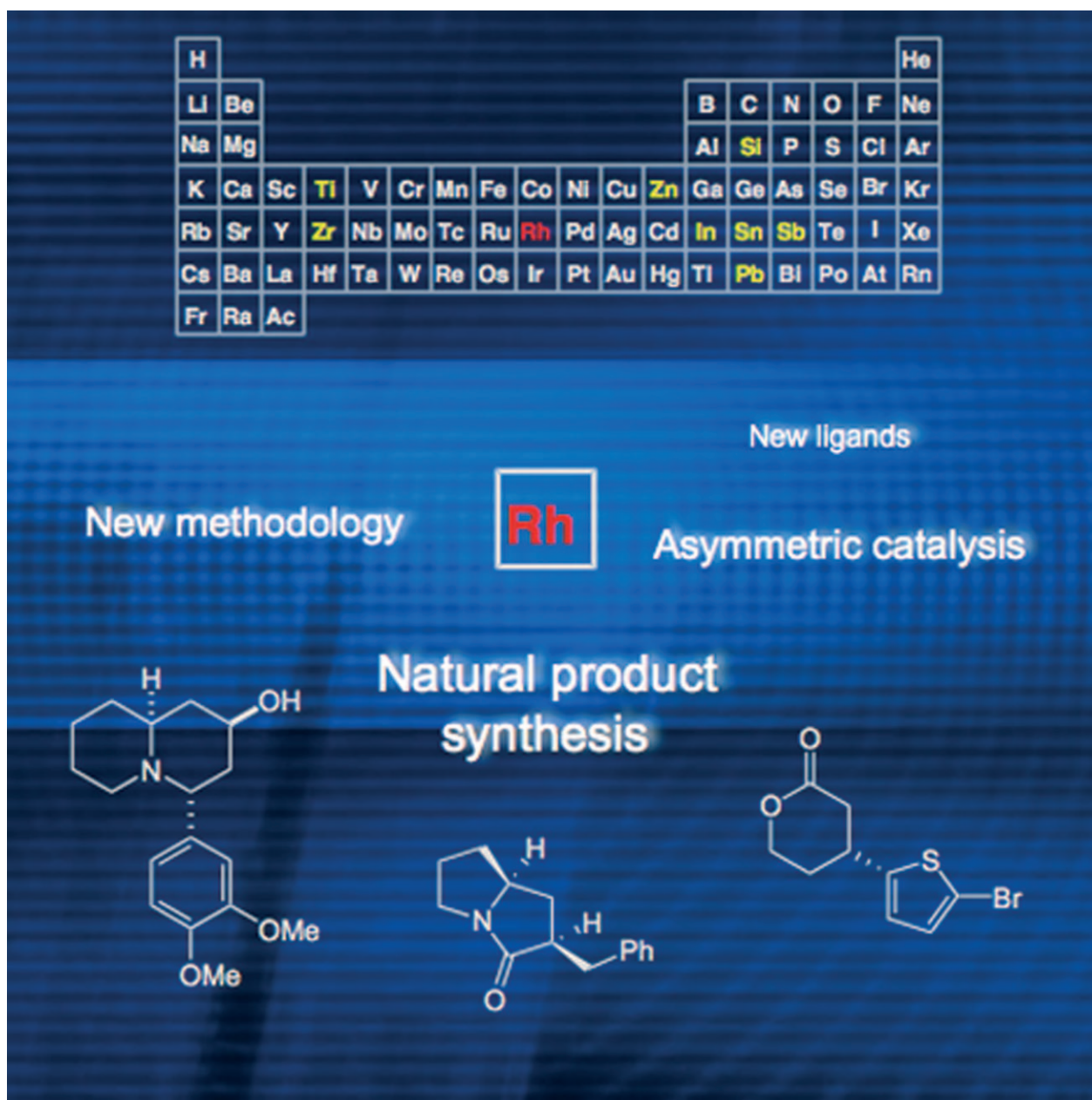


Alternatives to Organoboron Reagents in Rhodium-Catalyzed Conjugate Additions

Jonathan D. Hargrave, Joseph C. Allen, and Christopher G. Frost*^[a]

Dedicated to the 150th anniversary of Japan–UK diplomatic relations



Abstract: The rhodium-catalyzed conjugate addition of organoboron reagents to alkene acceptors has emerged as an important methodology in organic synthesis. The process results in the formation of new carbon–carbon bonds, and in principle, can create new stereocenters at both the α and β carbon atoms of the acceptor. Other organometallics are known to participate in the key transmetalation to rhodium and are becoming more established in stereo-

selective synthesis (notably arylzinc reagents). This Focus Review will describe recent developments in the use of alternative organometallic donors to organoboron reagents in rhodium-catalyzed conjugate addition reactions and their growing applications in synthesis.

Keywords: catalysis • conjugate addition • organosilicon reagents • organozinc reagents • rhodium

Introduction

The transition-metal-catalyzed conjugate addition of organometallics to activated alkenes is regarded as fundamental methodology for organic synthesis.^[1] Of particular importance is the elegant rhodium-catalyzed addition of organoboron reagents to α,β -unsaturated carbonyl derivatives, pioneered by Hayashi and Miyauchi.^[2] This widely used reaction can be carried out in aqueous solvent and affords excellent enantioselectivities (>90% *ee*) across a wide-range of substrates including α,β -unsaturated ketones, esters, amides, phosphonates, and sulfones. A number of excellent reviews have been published on this topic.^[3] A successful catalytic conjugate addition is dependent on an efficient transmetalation to rhodium followed by carbometallation to afford an η^3 -oxa- π -allylrhodium complex that is protonated to afford the product. Although often overlooked, a significant side reaction is the protodeboronation pathway that necessitates the use of two to five equivalents of organoboron donor to achieve satisfactory yields of product. The use of alternative organometallic donors to organoboron reagents can offer significant advantages in terms of intrinsic reactivity and selectivity and result in new transformations for organic synthesis based on rhodium-catalyzed conjugate addition.

1. Organozinc Reagents

Among the alternative organometallic species used in conjugate addition reactions, arylzinc chloride derivatives are

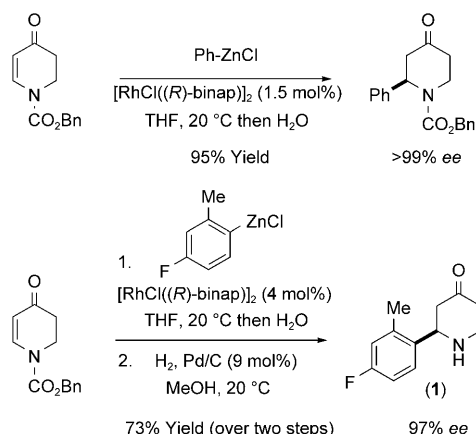
gaining popularity. They present the advantages of an easy preparation from the corresponding aryl halide compound and, thus, access to a wide range of substituted aryl fragments suitable for conjugate addition reaction. It is useful to note that water is not required for catalyst turnover when organozinc reagents are employed in rhodium-catalyzed conjugate addition reactions; transmetalation occurs at room temperature (or lower) and significantly, only 1.1 equivalents of the organometallic are normally required. In the first reported example, Hayashi and co-workers describe a rhodium-catalyzed conjugate addition reaction of organozinc reagents to prepare synthetically and biologically useful 2-aryl-4-piperidones.^[4] Initial investigations showed that the use of other organometallics such as PhB(OH)_2 gave lower conversions and yields of isolated product. PhZnCl , however, dramatically improved the level of reactivity under mild reaction conditions to afford arylated products in high yields and excellent selectivity (Scheme 1).

The reaction tolerates a diverse range of functionalized arylzinc reagents to give numerous arylated piperidones in excellent enantioselectivities. The potential for the methodology was demonstrated by the straightforward preparation of a key intermediate of tachykinin antagonists **1**.

Tandem catalytic reactions have emerged as valuable tools for efficient organic synthesis including enantioselective processes.^[5] In this context, the zinc enolates generated as the initial products from the conjugate addition of organozinc reagents can be further functionalized by the addition of electrophiles to the reaction mixture. This can be illustrated by the addition of pivaloyl chloride to afford O-acylated products in excellent yields or the addition of allyl bromide to provide α -allylated products as a single diastereomer (Scheme 2).

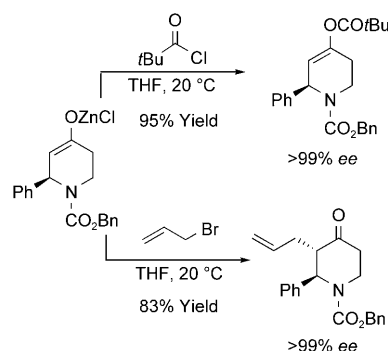
The rhodium-catalyzed asymmetric synthesis of 2-aryl-2,3-dihydro-4-quinolones via the addition of arylzinc reagents to 4-quinolones has been reported by Hayashi and co-workers

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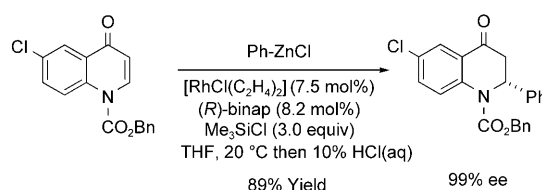


Scheme 1. Enantioselective synthesis of 2-aryl-4-piperidones.

as shown in Scheme 3.^[6] This report once again highlights the superior reactivity of arylzinc reagents against other organometallics, and demonstrated the excellent stereocontrol



Scheme 2. Sequential functionalization of zinc enolate.



Scheme 3. Enantioselective synthesis of 2-aryl-2,3-dihydro-4-quinolones.



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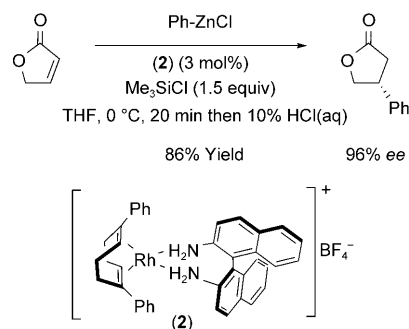
Jonathan D. Hargrave received his MChem (Hons) degree from the University of Bath in 2004. In 2008 he was awarded a PhD in organic chemistry with a thesis entitled 'Applications of rhodium catalysed 1,4-addition reactions in organic synthesis' under the direction of Dr. Christopher G. Frost. He currently occupies a research position within the pharmaceutical company Steris, focusing on the development of anticancer drug targets.



Christopher G. Frost completed his BSc (Hons) and PhD (with Jonathan M. J. Williams) at the University of Loughborough. After a postdoctoral fellowship with Prof. Philip D. Magnus at the University of Texas at Austin, he joined the Chemistry Department at the University of Bath and in 2007 was promoted to Reader in Organic Chemistry. His research interests are in the application of transition-metal catalysis to organic synthesis.

of (*R*)-binap to give products with high enantioselectivity in the presence of chlorotrimethylsilane. The resulting *R* configuration of the products from reactions employing Rh/(*R*)-binap was postulated to be due to the α -*Re*-face approach of the substrate avoiding steric repulsion between the phenyl group on the phosphorus atom of (*R*)-binap and the fused benzene ring of the substrate.

In the rhodium-catalyzed conjugate addition of organoboron reagents to alkene acceptors, chiral diene ligands can substitute for chiral phosphines in enantioselective processes and often show superior reactivity and enantioselectivity.^[7] The enantioselective synthesis of chiral dienes is challenging and the diversity of available ligands is fairly limited. However, Grützmacher and co-workers have reported an alternative approach to the synthesis of chiral diene-rhodium complexes where the diene is not chiral but prochiral until its coordination to a metal.^[8] In their approach, the optical resolution of a mixture of diastereomeric rhodium complexes coordinated with a diene and (*R*)-1,1'-binaphthyl-2,2'-diamine is readily achieved. Hayashi and co-workers have utilized the same strategy to prepare an enantiopure rhodium complex coordinated with 1,5-diphenyl-1,5-cyclooctadiene (Ph-cod). The complex [Rh((1*R*,5*R*)-Ph-cod)((*R*)-1,1'-binaphthyl-2,2'-diamine)]BF₄ (**2**) showed high catalytic activity and enantioselectivity in the addition of phenylzinc chloride to α,β -unsaturated ketones and esters in the presence of chlorotrimethylsilane as illustrated in Scheme 4.^[9] Interestingly, **2** was not an effective enantioselective catalyst for the conjugate addition of arylboronic acids. This is attributed to racemization of the catalyst under the reaction conditions (6 h at 50 °C), which could occur by dissociation of the diene from rhodium and recoordination on the other enantioface. By comparison, the catalytic addition of arylzinc



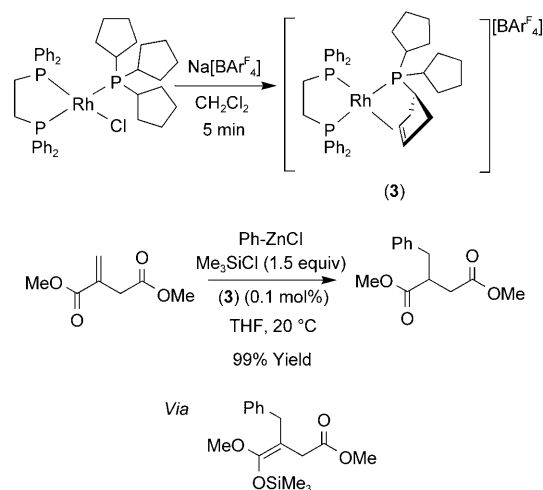
Scheme 4. New rhodium-diene complexes for enantioselective conjugate addition.

donors is extremely rapid (20 min at 0 °C) and the high enantioselectivity obtained suggests the reaction is complete before racemization of the catalyst becomes an issue.

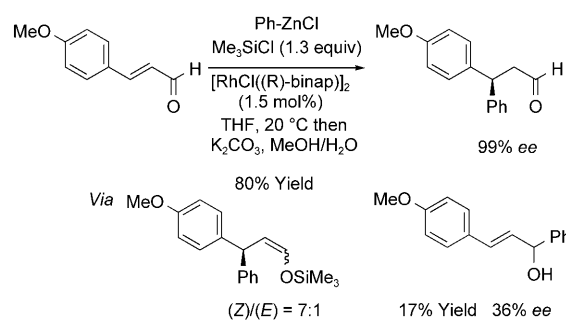
In previous studies, Hayashi and co-workers have established that hybrid phosphine-olefin ligands can offer distinct advantages over bidentate phosphine or diene ligands in the rhodium-catalyzed conjugate addition reaction.^[10] However, the synthesis of such ligands is multistep and can be nontrivial. A straightforward and facile synthesis of a rhodium complex containing a phosphine-olefin ligand has been reported by Weller and Frost.^[11] The rhodium complex (3) is assembled by the dehydrogenation of a cyclopentyl group in the commercially available cyclopentylphosphine (PCyp₃) ligand. Remarkably, this dehydrogenation reaction occurs rapidly (minutes) at room temperature without a hydrogen acceptor. The complexes are excellent catalysts for the conjugate addition of organometallics. In the addition of phenylzinc chloride to dimethyl itaconate, the reaction was complete after 2 h at room temperature with 1 mol % of 3 and within 16 h in the presence of 0.1 mol % of 3 (Scheme 5). That the active catalyst species is available in excellent yields and purity from readily available starting materials in two simple steps makes this methodology particularly attractive.

An elegant example of the chemoselective asymmetric conjugate addition of arylzinc chlorides to (*E*)-3-arylpropenals has been reported by Hayashi and co-workers (Scheme 6). The reactions proceed with high enantioselectivity in the presence of a Rh/(*R*)-binap complex to yield 3,3-diarylpropanals after hydrolysis of the intermediate silyl enol ether (consisting of *Z* and *E* isomers in a ratio of 7:1).^[12] Interestingly it was shown that chlorosilane was essential for high yields of the conjugate addition product with only minor amounts of 1,2-addition observed.

Despite the advances in catalyst design and substrate (acceptor) diversity, the structure of the organometallic donor remains surprisingly limited to aryl and a small number of alkenyl derivatives (*alkyl* organometallics are not currently viable donors as the rhodium alkyl complexes are kinetically unstable and rapidly undergo β-hydride elimination). In a research program intended to expand the practical utility of the rhodium-catalyzed conjugate addition reactions, Frost



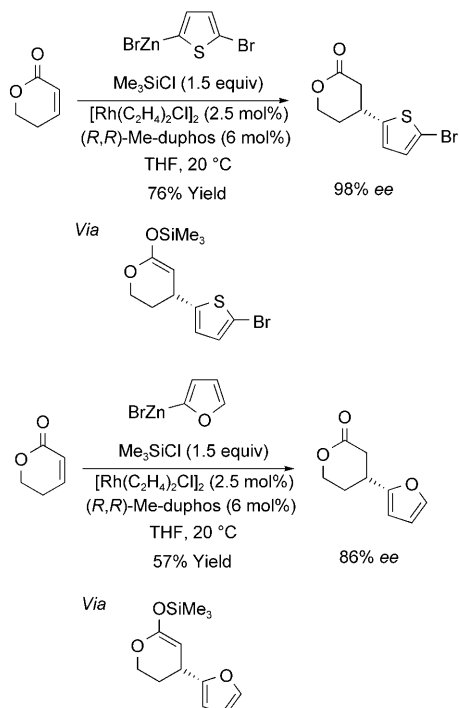
Scheme 5. Synthesis and catalytic application of new phosphine-olefin ligands.


 Scheme 6. Enantioselective synthesis of (*E*)-3-arylpropenals.

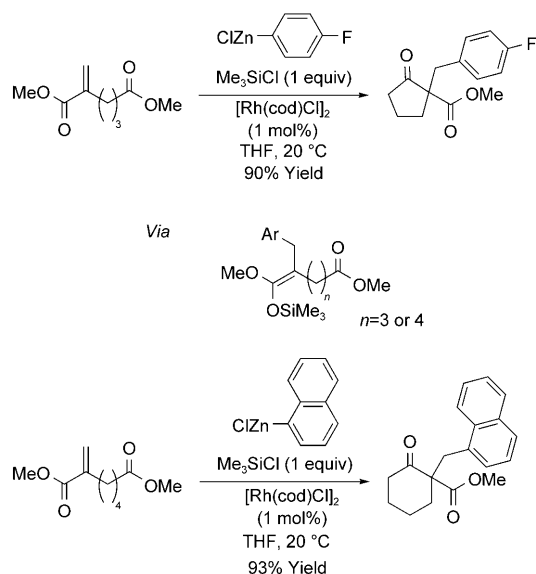
and co-workers have successfully demonstrated that 2-heteroarylzinc donors can be utilized in a highly enantioselective process.^[13] A complex derived from [RhCl(C₂H₄)₂]₂ and Me-duphos is reported to be superior to established catalyst systems for this challenging transformation. A range of 2-heteroarylzinc donors were noted to undergo highly enantioselective additions as illustrated in Scheme 7 for the cyclic lactone 5,6-dihydro-2*H*-pyran-2-one.

Tandem catalytic processes triggered by conjugate addition offer significant potential in synthesis.^[14] In this context, Frost and co-workers have demonstrated a new method for constructing quaternary carbon centers via a tandem rhodium-catalyzed conjugate addition/intramolecular cyclisation reaction (Scheme 8).^[15] The use of highly reactive arylzinc halides as the organometallic donors was essential to the efficiency of the tandem process. The transformation led to the facile preparation of substituted cyclopentanones and cyclohexanones containing quaternary carbon centers under mild conditions and in high yields from acyclic precursors.

Hayashi and co-workers showed that the rhodium-catalyzed addition of arylzinc reagents to aryl alkynyl ketones was possible, giving highly substituted indanones.^[16] In this reaction, the choice of the dppf ligand was crucial in dictat-

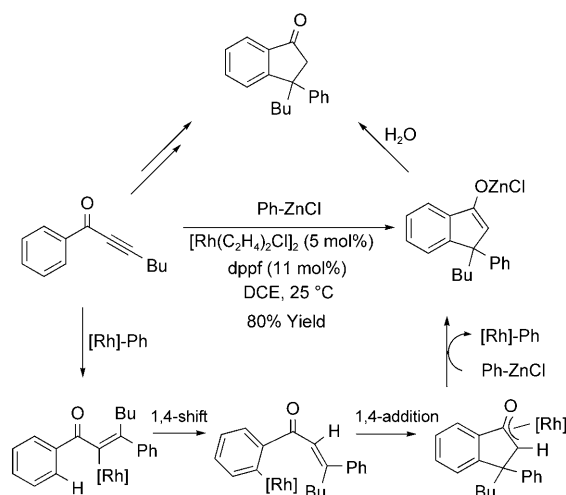


Scheme 7. Application of heteroaromatic donors.



Scheme 8. Tandem addition/intramolecular cyclization reaction.

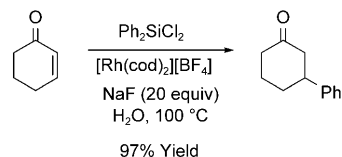
ing the reaction pathway. The use of bisphosphine ligands other than dppe produced only a negligible amount of the desired indanone, and instead other arylated products were isolated. The proposed reaction pathway is shown in Scheme 9 and is thought to proceed via a 1,4-shift of an organorhodium species followed by a 1,4-addition. The zinc enolate generated is quenched with water to afford substituted indanones in good yields.



Scheme 9. Addition of arylzinc reagents to aryl alkynyl ketones.

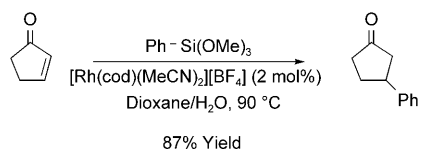
2. Organosilicon Reagents

The utility of arylchlorosilanes in rhodium-catalyzed conjugate additions has been reported by Li and Huang. For example, the addition of diphenyldichlorosilane to cyclohexenone is achieved in the presence of a high loading of cationic rhodium catalyst.^[17] The reaction proceeds in water as solvent but requires the addition of a large excess of fluoride salt for an efficient conversion to product (Scheme 10).



Scheme 10. Rhodium-catalyzed addition of arylchlorosilanes.

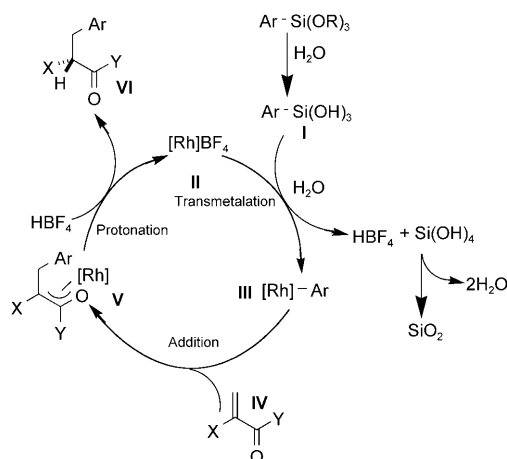
Oi and Inoue have reported the additive-free conjugate addition of organotrialkoxysilanes.^[18] The addition of phenyltrimethoxysilane to cyclopentenone in the presence of just 2 mol % of cationic rhodium complex furnishes the product in good yield (Scheme 11). The cationic rhodium complex $[\text{Rh}(\text{cod})_2][\text{BF}_4]$ successfully catalyzes the 1,4-addition of organotrialkoxysilanes to α -substituted acrylic esters. The reactions were shown to be effectively enhanced by microwave irradiation leading to a range of functionalized 2-alkyl succinates and α -amino acid derivatives in as little as



Scheme 11. Rhodium-catalyzed addition of organotrialkoxysilanes.

50 min.^[19] The use of organotrialkoxysilanes offer many of the advantages of boronic acids in terms of commercial availability, ease of handling, and stability to air and moisture. Furthermore, a number of straightforward synthetic routes have been reported for their preparation including the hydrosilylation of alkynes, the addition of Grignard reagents to tetraethyl orthosilicate, and the cross-coupling of triethoxysilane with organohalides.^[20]

The mechanism of the process as proposed by Oi and Inoue is illustrated in Scheme 12 and closely relates to the mechanism of the corresponding rhodium-catalyzed addition

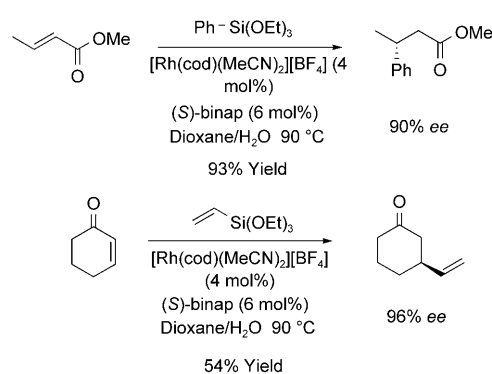


Scheme 12. Mechanism of organotrialkoxysilane addition.

of boronic acids described by Hayashi.^[21] It was noted that the reaction did not occur in the absence of water, suggesting that the active silicon species is likely to be an organosilanetriol **I** generated from the hydrolysis of the organotrialkoxysilane. The organosilanetriol **I** would then transmetalate with the cationic catalyst **II** to generate the organorhodium intermediate **III** and the side product SiO_2 . Addition of the organorhodium species to the substrate **IV** then leads to an η^3 -oxa- π -allylrhodium complex **V** that is protonated to afford the product **VI** and the regenerated cationic catalyst. Alternatively, the lack of reactivity in the absence of proton source could be attributed to the η^3 -oxa- π -allylrhodium complex **V** being inert to transmetalation, which would lead to the catalytic cycle stalling at **V**.

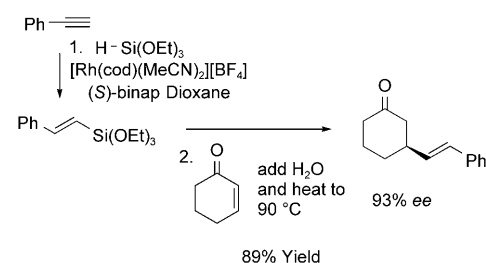
More recently, the rhodium-catalyzed enantioselective conjugate addition of organotrialkoxysilanes to α,β -unsaturated carbonyl compounds has been disclosed.^[22] Oi and co-workers have demonstrated that enantiopure cationic rhodium complexes can promote the addition of organotrialkoxysilanes to α,β -unsaturated ketones, esters, and amides.^[23] The methodology utilizes binap as a chiral ligand and allows the introduction of aryl or alkenyl groups at the β -position with high enantioselectivity (Scheme 13).

Hayashi and co-workers noted that the cationic rhodium complex $[\text{Rh}((S)\text{-binap})(\text{MeCN})_2]\text{BF}_4$ used for conjugate additions of organotrialkoxysilanes to enones, also catalyzes the hydrosilylation of acetylenes with triethoxysilane.^[24] In



Scheme 13. Enantioselective conjugate addition of organotrialkoxysilanes.

light of this discovery it was demonstrated that a one-pot procedure could be carried out whereby an enantiopure rhodium complex can catalyze two successive reactions to yield alkenyl-substituted cyclic ketones (Scheme 14). Thus, phe-

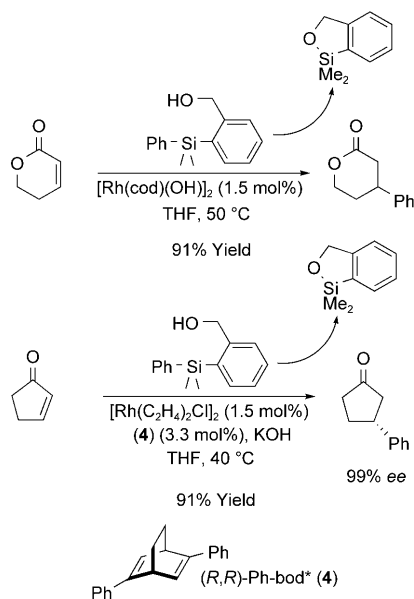


Scheme 14. One-pot rhodium-catalyzed hydrosilylation/conjugate addition.

nylacetylene is treated with triethoxysilane in the presence of $[\text{Rh}((S)\text{-binap})(\text{MeCN})_2][\text{BF}_4]$ and $(S)\text{-binap}$ in dioxane. After one hour at room temperature, water and 2-cyclohexenone are added and the mixture is heated to 90°C for 20 h. The product $(S)\text{-3-}((E)\text{-2-phenylethenyl})\text{cyclohexanone}$ is obtained in good yield and with high enantioselectivity as the *Z* isomer in a ratio of 99:1.

Hayashi and co-workers have introduced organo[2-(hydroxymethyl)phenyl]-dimethylsilanes as practical reagents for conjugate addition reactions to α,β -unsaturated carbonyl acceptors.^[25] The ready accessibility and high stability towards acid, base, and moisture establish the new silane reagents as an attractive alternative to other organometallic donors. The catalytic protocol tolerates a diverse range of functional groups and is applicable to gram-scale synthesis. The use of a chiral diene ligand such as $(1R,4R)\text{-2,5-diphenylbicyclo[2.2.2]octa-2,5-diene}$ (*R,R*)-Ph-bod* (**4**) enables an efficient and reliable enantioselective transformation (Scheme 15).

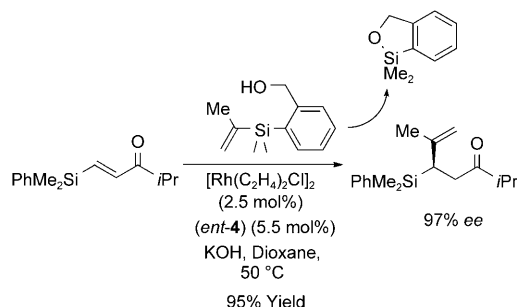
The authors have expanded this methodology to include the conjugate addition of alkenyl[2-(hydroxyl)phenyl]dimethylsilanes to β -silyl- α,β -unsaturated ketones.^[26] With the application of $(S,S)\text{-Ph-bod}^*$ (*ent*-**4**) as a ligand, a range of al-



Scheme 15. Organo[2-(hydroxymethyl)phenyl]-dimethylsilanes in conjugate additions.

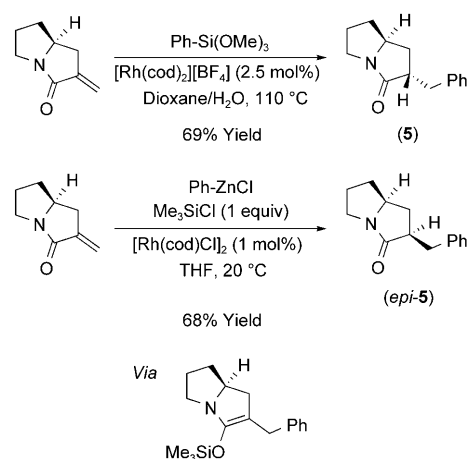
kenyl nucleophiles have been installed to these substrates in high yields and enantiomeric excess (Scheme 16).

In the majority of reported rhodium-catalyzed conjugate additions to prochiral acceptors, a single stereocenter is es-



Scheme 16. Organo[2-(hydroxymethyl)phenyl]-dimethylsilanes in conjugate additions.

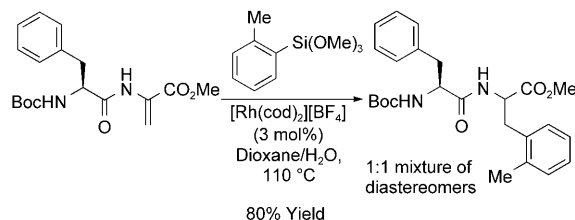
ablished by an asymmetric carbometallation under control of an enantiopure rhodium complex. Interestingly, despite the tremendous synthetic potential, there are relatively few examples of rhodium-catalyzed conjugate additions to enantiopure acceptors. In this context, Frost and co-workers have reported a convenient stereoselective route to functionalized pyrrolizidinones utilizing the rhodium-catalyzed conjugate addition reaction.^[27] Interestingly it was noted that a switch in stereoselectivity occurred when changing the organometallic reagent from arylzincs to aryltrialkoxysilanes. As illustrated in Scheme 17, the reaction proceeded to afford the (2*S*,7*S*) stereoisomer (**5**) as the major product with phenyl-trimethoxysilane and the (2*R*,7*S*) stereoisomer (*epi*-**5**) as the major product with phenylzinc chloride as donor. This switch was postulated to be attributed to the rate of trans-



Scheme 17. Stereoselective synthesis of functionalized pyrrolizidinones.

metalation, which dictates the reaction temperature and subsequent mode of protonation. Importantly, *epi*-**5** is formed by hydrolysis of the silyl enol ether.

In related studies, the rhodium-catalyzed conjugate addition of aryltrialkoxysilanes to α,β -dehydroamino acids enables a site-selective modification of peptides.^[28] The Boc-Phe- Δ Ala-OMe dipeptide was prepared from the corresponding serine precursor by a β -elimination process using EDCI and Cu^ICl. The addition proceeded efficiently to afford the product as a mixture of diastereomers (Scheme 18).

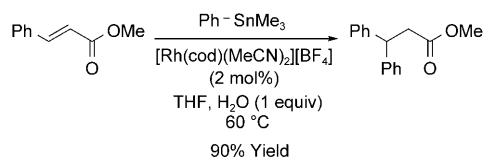


Scheme 18. Modification of peptides.

3. Organotin Reagents

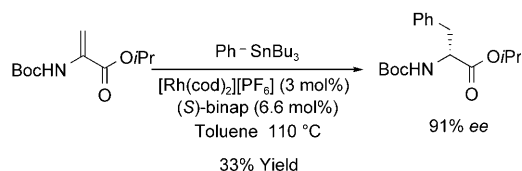
Inoue and Oi first reported that rhodium complexes could promote the conjugate addition of organostannanes to α,β -unsaturated compounds under anhydrous conditions.^[29] Subsequently, Li and co-workers noted that the rhodium-catalyzed addition of trialkylphenyl and trialkylvinyltin reagents proceeded in aqueous media to afford the conjugate addition products under an air atmosphere.^[30] Interestingly, the reaction shuts down when the alkyl and aryl substituents on the tin center were switched to halogens such as bromo and chloro but commenced again when a base was added. The transient formation of organotin hydroxide intermediates which demonstrated enhanced reactivity of the C–Sn bonds in the transmetalation step was proposed to account for this observation.^[31] Further work by Inoue and Oi revealed the

cationic catalyst $[\text{Rh}(\text{cod})(\text{MeCN})_2]\text{BF}_4$ afforded conjugate addition products in high yields for a range of unsaturated ketones (Scheme 19). It was observed that the addition of water to the reaction was key to obtaining high yields.^[32]



Scheme 19. Addition of organostannanes in aqueous media.

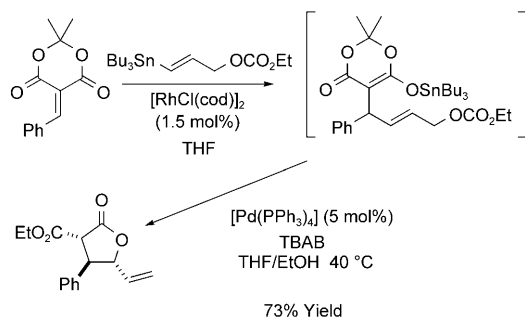
Genet and Darses have reported a remarkable enantioselective synthesis of α -amino acids featuring the rhodium-catalyzed conjugate addition of phenylstannanes to dehydroalanine derivatives.^[33] In the presence of $[\text{Rh}(\text{cod})_2][\text{PF}_6]$ and (*S*)-binap, the addition of phenyltributylstannane occurs to afford a modest yield of product with excellent enantioselectivity (Scheme 20). By contrast, organoboron reagents do



Scheme 20. Enantioselective synthesis of amino acids.

not afford any of the expected product in the absence of an added base or proton source. The proposed mechanism implies a conjugate addition type reaction of the organometallic reagent followed by an intramolecular hydride transfer from nitrogen to the adjacent carbon.

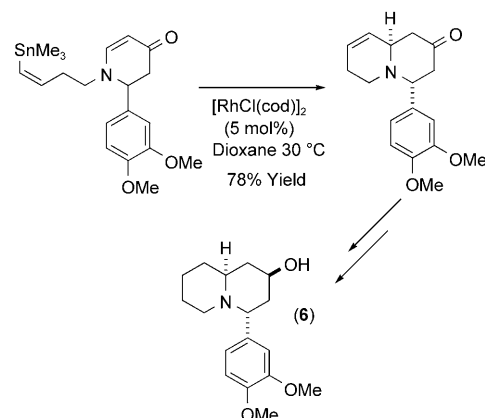
Fillion and co-workers have developed ambiphilic reagents based on 3-(tributylstannyl)-2-propenyl acetates and carbonates. The vinyl–tin bond can be used as a nucleophile in rhodium-catalyzed conjugate additions and the allyl carbonate as a palladium-activated electrophile.^[34] This is illustrated in the addition of functionalized vinyltin reagents to alkylidene Meldrum's acids, followed by palladium-catalyzed intramolecular allylation to provide direct entry into



Scheme 21. Sequential catalytic synthesis of vinyl-substituted γ -lactones.

vinyl-substituted γ -lactones via O-alkylation (Scheme 21) and vinylcyclopropanes via C-alkylation.

Furman and co-workers have developed an elegant rhodium-catalyzed intramolecular addition of vinylstannanes to dihydro-4-pyridones providing a stereoselective route to 1-azabicyclic alkaloids.^[35] As depicted in Scheme 22, this methodology was applied to a concise synthesis of racemic Lasubine I (**6**), an interesting quinolizidine alkaloid from the plants of the Lythraceae family.



Scheme 22. Stereoselective synthesis of 1-azabicyclic alkaloids.

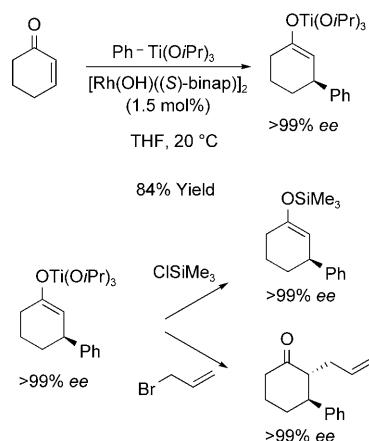
4. Organotitanium Reagents

Hayashi and co-workers have developed the rhodium-catalyzed asymmetric conjugate addition of aryltitanium triisopropoxides ($\text{ArTi}(\text{OiPr})_3$) to furnish enantioenriched titanium enolates with high enantioselectivities.^[36] The generated enantioenriched titanium enolates can be utilized in a range of subsequent reactions including involving the formation of isolable silyl enol ethers or functionalization in the α -position with a range of electrophiles (Scheme 23).

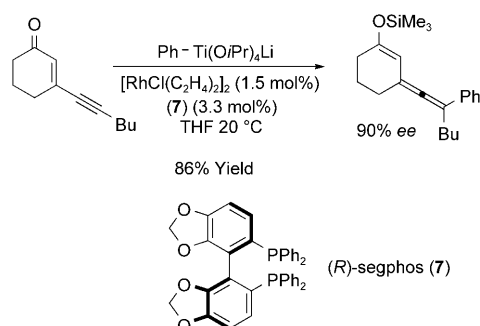
The addition of aryltitanate reagents has also been performed on 3-alkynyl-2-1-ones in the presence of chlorotri-methylsilane and a rhodium-(*R*)-segphos complex (Scheme 24). This reaction proceeds in a 1,6-fashion to afford enantioenriched allenylalkenyl silyl enol ethers in high yields and with up to 93 % *ee*. Although the allenes were isolated as silyl enol ethers, they could also be converted into enol pivalate esters or enol triflates.^[37]

5. Organozirconium Reagents

Oi, Sato, and Inoue have demonstrated the elegant enantioselective addition of alkenylzirconocene chlorides to cyclic or acyclic enones. This is an attractive method as the alkenylzirconium intermediate can be formed in situ by reaction of a terminal alkyne with chloridobis(η^5 -cyclopentadienyl)-hydrido-zirconium (Schwartz reagent). The authors observed that high enantioselectivity could be obtained by using an

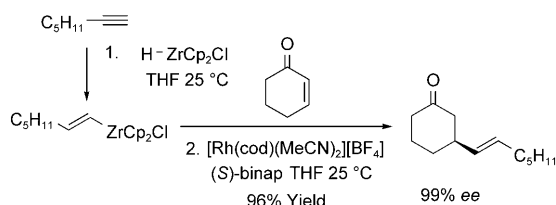


Scheme 23. Synthesis and functionalization of enantioenriched titanium enolates.



Scheme 24. Enantioselective 1,6-addition of aryltitanate reagents.

enantipure rhodium complex generated from $[\text{Rh}(\text{cod})-(\text{MeCN})_2]\text{BF}_4$ and $(S)\text{-binap}$ (Scheme 25).^[38]

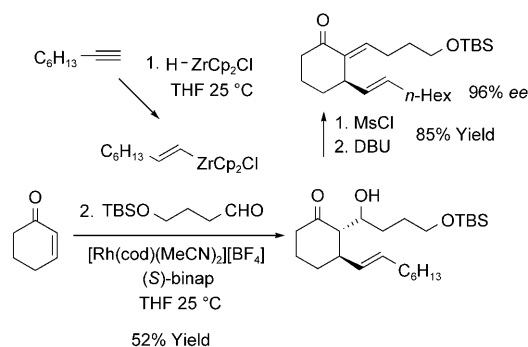


Scheme 25. Enantioselective addition of alkenylzirconocene chlorides.

Nicolaou and co-workers have taken this methodology to an extra level by developing a catalytic asymmetric three-component reaction as shown in Scheme 26. The products were used for the construction of the spirocyclic domain of the marine natural product vannusal A.^[39]

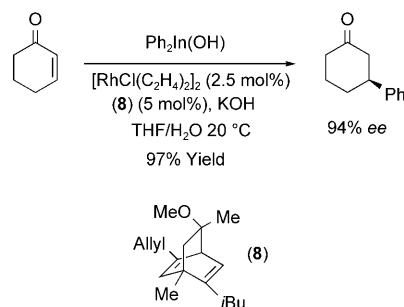
6. Other Main-Group Organometallics

Murakami and co-workers demonstrated the utility of diarylindium(III) hydroxides in rhodium-catalyzed conjugate addition reactions.^[40] The reagents are readily prepared



Scheme 26. Catalytic asymmetric three-component reaction.

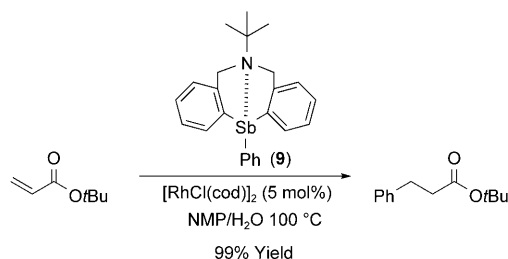
from the corresponding organolithium or organomagnesium reagent and used without further purification. The use of enantiopure diene ligand **8** resulted in the formation of arylated compounds in good yields and excellent enantioselectivity (Scheme 27).



Scheme 27. Enantioselective addition of diarylindium(III) hydroxides.

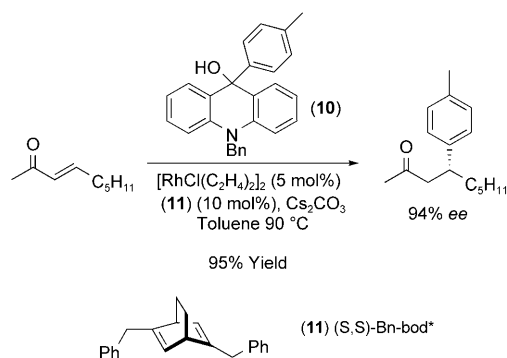
Li and co-workers have demonstrated the application of robust triphenylbismuth^[30] and phenyltrimethyllead reagents^[41] in rhodium-catalyzed conjugate additions. An interesting report reveals that organoantimony compounds and specifically 1,5-azastibocenes (**9**) with N-Sb intramolecular interaction are excellent transmetalating agents for rhodium-catalyzed conjugate addition reactions.^[42] This is illustrated in the efficient arylation of *tert*-butyl acrylate by **9** in the presence of $[\text{RhCl}(\text{cod})]_2$ (Scheme 28).

The generation of arylrhodium species from metal alkoxides through β -hydrocarbyl elimination is a promising strategy for catalytic synthesis. Hayashi and co-workers have de-



Scheme 28. Addition of 1,5-azastibocenes.

veloped a rhodium-catalyzed arylation of α,β -unsaturated carbonyl compounds by the use of trisubstituted aryl methanols.^[43] The optimum aryl transfer reagents were derived from 10-benzylacridin-9-(10H)-one by addition of the appropriate aryl Grignard reagent. The use of chiral diene ligand (*S,S*)-Bn-bod* (**11**) enables the efficient rhodium-catalyzed transfer of aryl group from **10** to the substrate in an enantioselective fashion (Scheme 29).



Scheme 29. Enantioselective conjugate addition using new aryl-transfer reagents.

Conclusions

The rhodium-catalyzed conjugate addition of organometallics to activated alkenes is a powerful synthetic tool for establishing new carbon–carbon bonds, often with high enantioselectivity. The advent of a practical, reliable method for introducing functionalized aryl and alkenyl fragments with predictable stereocontrol has caught the attention of the synthetic community and elegant applications are starting to emerge in the construction of natural products and the large-scale synthesis of intermediates in drug discovery. The utility of a broader range of organometallic donors has been demonstrated to offer opportunities in terms of reactivity, selectivity, and the discovery of new transformations for organic synthesis based on rhodium-catalyzed conjugate addition. The development of efficient tandem processes, the use of enantiopure acceptors, and the modification of complex substrates such as peptides represent innovative new methods in catalysis research facilitated by the rhodium-catalyzed conjugate addition reaction. The challenge for the future will be to offset the high cost of rhodium by increasing efficiency and using immobilized rhodium complexes to enable catalyst recovery and recycling.

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